

MAC Address, ARP & Spoofing

1. MAC Address

A **MAC (Media Access Control) address** is a 48-bit identifier associated with a network interface. It is commonly represented in hexadecimal notation, with the bytes separated by colons.

Example: 1A:3D:9F:33:94:21

In the traditional representation, the first 24 bits identify the manufacturer (OUI — Organizationally Unique Identifier), while the remaining 24 bits identify the network interface within that allocation.

- Operates at **Layer 2 (Data Link Layer)** of the OSI model.
- Helps identify network interfaces for communication on a local network.
- A device can have different MAC addresses for different network interfaces.
- A MAC address is independent of the device's IP address, so changing an IP address does not inherently change the MAC address.

2. MAC Spoofing

MAC spoofing is a technique used to change or override the MAC address presented by a network interface in software. Although a hardware interface may have a manufacturer-assigned address, an operating system or driver can sometimes configure a different address for network communication.

Why People Use MAC Spoofing

- **Privacy & Anonymity:** Randomizing a MAC address can reduce long-term device tracking on some public Wi-Fi networks.
- **Bypassing Restrictions:** Some networks use MAC filtering or device-based access controls; changing the presented MAC can bypass weak implementations of these controls.
- **Network Testing:** Developers and security professionals may use spoofed addresses to test network behavior and access controls.
- **Malicious Use:** Attackers may impersonate another device at the local network layer to attempt unauthorized access or facilitate attacks such as man-in-the-middle activity.

Defending Against MAC Spoofing

- **Port Security:** Switches can be configured to restrict which MAC addresses are permitted on specific ports and to react to unexpected addresses.
- **Monitoring:** Tools such as **arpwatch** can monitor changes in IP-to-MAC address relationships and alert administrators to unexpected changes.
- **802.1X Authentication:** Provides stronger access control through authenticated network access rather than relying only on MAC addresses.

3. ARP — Address Resolution Protocol

ARP (Address Resolution Protocol) maps an IPv4 address to a MAC address on a local network. It bridges the gap between Layer 3 logical addressing and Layer 2 Ethernet addressing.

ARP was introduced in **1982 through RFC 826** and is primarily used with IPv4 networks. IPv6 uses Neighbor Discovery Protocol (NDP) instead of ARP.

ARP Cache

After an ARP resolution, the learned IP-to-MAC mapping can be stored temporarily in an **ARP cache**. Subsequent communication can use the cached mapping instead of performing a new ARP request each time.

- **ARP Cache Timeout:** The period for which a learned ARP entry can remain in the cache before it expires or is refreshed.

View ARP information: `arp -a` or `ip neigh`

ARP Request

An ARP request is normally sent as a **broadcast** on the local network to discover which device owns a particular IPv4 address.

- Sender MAC address.
- Sender IPv4 address.
- Target IPv4 address being queried.
- The Ethernet destination for the request is the broadcast address: `FF:FF:FF:FF:FF:FF`.

ARP Reply

The device that owns the requested IPv4 address can respond with an ARP reply containing its MAC address. The requesting host can then store the IP-to-MAC mapping in its ARP cache and use it for subsequent local communication.

ARP Table

An ARP table stores mappings between IPv4 addresses and corresponding MAC addresses for neighboring devices. On Linux, `ip neigh` can display the neighbor table.

4. ARP Spoofing / ARP Poisoning

ARP spoofing, also called **ARP poisoning**, is an attack in which forged ARP messages are used to associate an attacker's MAC address with another device's IPv4 address. This can cause local traffic to be sent to the attacker instead of the intended host.

Possible Goals and Effects

- **Traffic interception:** The attacker may position themselves between two communicating hosts.
- **Spying:** The attacker may observe intercepted communications and forward them to the intended destination.
- **Man-in-the-Middle (MitM):** The attacker may intercept and potentially modify traffic before forwarding it.
- **Denial of Service:** Incorrect ARP mappings can disrupt communication between hosts.
- **Further Attacks:** Intercepted traffic can potentially be used as part of a broader attack chain.

How ARP Spoofing Works — Simplified

- The attacker sends forged ARP information on the local network.
- A victim associates the attacker's MAC address with a trusted IP address.
- Traffic intended for that IP may then be sent to the attacker.
- The attacker may observe, forward, modify, or drop traffic depending on the attacker's position and configuration.

5. Who Uses ARP Spoofing?

ARP spoofing can be used for both malicious and legitimate security purposes. Attackers may use it to intercept traffic or disrupt communications. Developers and ethical hackers may also simulate ARP cache poisoning in controlled environments to test network defenses.

6. Effects of ARP Spoofing

- The attack may initially produce few obvious symptoms.
- Network traffic may be redirected through an unexpected device.
- Communications may be observed or modified.
- Connections can become unstable or fail completely if traffic is disrupted.
- ARP poisoning can be used as a step toward additional attacks.

The visible effects depend on the attacker's objective and whether the attack is being used for interception, modification, disruption, or as part of a larger attack.

7. Protection Against ARP Spoofing

Network defenses can reduce the risk and impact of ARP spoofing.

- **Packet Filtering / Network Controls:** Security devices and switches can apply controls that detect or restrict suspicious traffic and inconsistent address mappings.
- **Dynamic ARP Inspection (DAI):** Managed switches can validate ARP messages against trusted IP-to-MAC bindings.
- **DHCP Snooping:** Can help build trusted IP-to-MAC bindings that other switch security features can use.
- **Static ARP Entries:** In appropriate small or controlled environments, static mappings can reduce reliance on dynamic ARP.
- **Encryption:** HTTPS, TLS, VPNs, and other encrypted protocols help protect the confidentiality and integrity of data even if traffic is intercepted.
- **Network Segmentation:** Proper segmentation can limit the scope of local-layer attacks.
- **Monitoring:** Monitor ARP/neighbor-table changes and investigate unexpected IP-to-MAC relationships.

Important Security Note

Encryption does not prevent ARP spoofing itself. Instead, strong encryption can reduce what an attacker can learn or modify from intercepted traffic. For example, HTTPS protects web traffic from being read as plaintext by a network intermediary, although it does not make the underlying network attack harmless.

8. MAC Spoofing vs ARP Spoofing

Aspect	MAC Spoofing	ARP Spoofing
What changes?	The MAC address presented by a network interface	ARP mappings between IPv4 addresses and MAC addresses
Primary scope	Local network interface identity	Local IPv4 neighbor/address resolution
Main purpose	Privacy, testing, bypassing weak MAC-based controls, or impersonation	Traffic interception, MitM activity, or disruption
Typical target	Network access controls or device identity checks	Victim ARP caches / local network traffic
Key defenses	802.1X, port security, monitoring	DAI, DHCP snooping, segmentation, encryption, monitoring

9. Cybersecurity Relevance

- MAC addresses help analysts understand Layer 2 communication and local network behavior.
 - ARP tables and packet captures can reveal IP-to-MAC relationships.
 - MAC spoofing demonstrates why MAC addresses alone should not be treated as strong authentication.
 - ARP spoofing is an important concept in network monitoring, penetration testing, and incident response.
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- Wireshark and command-line networking tools can help analyze MAC and ARP activity.

Key Takeaway

MAC Address: A Layer 2 identifier associated with a network interface.

ARP: Resolves IPv4 addresses to MAC addresses on a local network.

MAC Spoofing: Changes the MAC address presented by an interface.

ARP Spoofing: Manipulates IP-to-MAC mappings to redirect or disrupt local traffic.

Memory aid: MAC identifies the local interface → ARP finds the MAC for an IPv4 address → ARP spoofing abuses that resolution process.